Extraction of Aluminium from Bauxite

Exam code: 0620 & 0971

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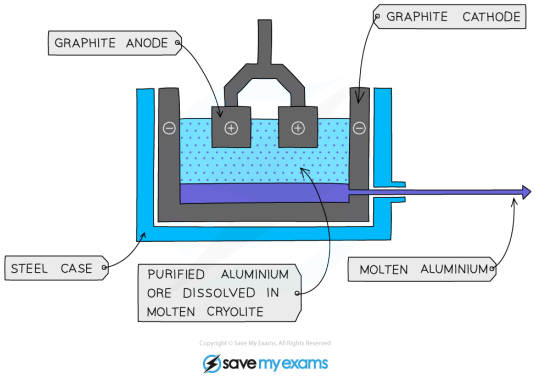
Help me understand

Extraction of aluminium from bauxite

- Aluminium is a reactive metal, above carbon in the reactivity series
- Its main ore, is **bauxite**, which contains aluminium oxide
- Aluminium is higher in the reactivity series than carbon, so it cannot be extracted by reduction using carbon
- Instead, aluminium is extracted by **electrolysis**

The electrolysis of aluminium



Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Examiner Tips and Tricks

If you are a Core student, you do **not** need to explain the process of extraction of aluminium by electrolysis.

The process of aluminium extraction by electrolysis

Extended tier only

- Bauxite is first **purified** to produce aluminium oxide, Al_2O_3
- Aluminium oxide is then dissolved in **molten cryolite**
  - This is because aluminium oxide has a melting point of over 2000 °C which would use a lot of energy and be very expensive
  - The resulting mixture has a lower melting point without interfering with the reaction
- The mixture is placed in an electrolysis cell, made from steel, lined with graphite
- The **graphite** lining acts as the negative electrode, with several large graphite blocks as the positive electrodes
- At the **cathode** (negative electrode)**:**
  - Aluminium ions gain electrons (reduction)
  - Molten aluminium forms at the bottom of the cell
  - The molten aluminium is siphoned off from time to time and fresh aluminium oxide is added to the cell

$$Al^{3+} + 3e^- \rightarrow Al$$

- At the **anode** (positive electrode)**:**
  - Oxide ions lose electrons (oxidation)
  - Oxygen is produced at the anode:

$$2O^{2-} \rightarrow O_2 + 4e^-$$

- The overall equation for the reaction is:

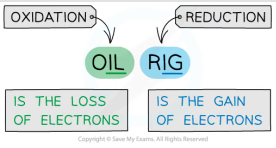
$$2Al_2O_3 \rightarrow 4Al + 3O_2$$

- The carbon in the **graphite** anodes reacts with the oxygen produced to produce $\mathbf{CO_2}$

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

- As a result the anode wears away and has to be replaced regularly
- A lot of **electricity** is required for this process of extraction, this is a major **expense**

Examiner Tips and Tricks



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